PATTERN_FIBRE_RR: 140-label pooled $5u^2/2$ mass mod 8 vs MF2

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RESULTS: PATTERN_FIBRE_RR (Map-PFR1)

PREREG sha256 8b00bced504370a7ea0c9baa1e15e82af33aa065dd7127f848cafcdfccf3331a. Hashed after explore (data/explore.sha256) and before test (logs/test.log, data/test.json). test.py recomputes the label set independently and matches explore.

fibre	ρ_res (MF2)	ρ_mod (140-label pool, $(10u)^2$ mod 8)
0	1/32	1/4
1	1/4	1/2
2	3/32	0
3	0	0
4	7/32	1/4
5	1/4	0
6	5/32	0
7	0	0

- P1 (exact equality): false. TV = 1/2.
- Supports: {0,1,4} vs {0,1,2,4,5,6}.
- Zeros on {3,7}: yes, but this is a consequence of the square-class obstruction. 2, 5 and 6 vanish too.
- Non-vacuous: ρ_mod is not uniform on the six open fibres.

Verdict: FAIL (Category A, exact rationals). Bridge: **NO.** ## NOTES: PATTERN_FIBRE_RR ## Explore (data/explore.json) - Written label family reproduced: 70 α's × 2 β's = 140 labels. The α1 values are {0, 1/10, 1/5, 3/10, 2/5, 1/2, 3/5, 7/10, 4/5, 9/10}. - Cross-checks against the text: (7/10,4/5) and (1/2,j/7) are present, and exactly 28 β=−a labels have 70α1 ∈ {7,21,49,63} (rev8 l.786–787: "gives 28 labels"). - Minimal integral scale N = 40 (the denominators of α1 divide 10). The ×68 convention is not integral here, so it is not used. - Per α1 (N=40): - α1 ∈ (1/5)Z gives (1/2,0,0,0,1/2,0,0,0). - α1 ∈ 1/10 + (1/5)Z gives mass 1 on fibre 1. - Pools: - all 140 labels: (1/4,1/2,0,0,1/4,0,0,0) (COMMITTED); - 28 labels with 70α1 ∈ {7,21,49,63}, β=−a: (0,1,0,0,0,0,0,0). - The paper's condition 2α1 ∈ Z holds only for α1 ∈ {0,1/2}; that subset gives fibres {0,4} and {1}.

Lemma PF1 (Category A): pooled square-class obstruction

Name: PF, not L, since L1–L7 are Einstein–Hampton only. Fibre lemmas elsewhere are MF1–MF3. For any label pool, any weights and any common integral scale N, the exponent is $N \cdot 5u^2/2 = (N/40) \cdot x^2$ with $x = 10u \in Z$. So every fibre lies in $c \cdot \{0,1,4\}$ mod 8 (at most 3 classes). ρ_{res} has 6 classes. So no pooling of the single unary theta $5u^2/2$ over the written label family can give MF2. This holds for the unshifted theta (BATCH8) and for every label shift.

Next pattern step (not run)

The obstruction is in the unary exponent itself. The written object that escapes it is the **binary** form of the same family: $Q(v) = v1^2 + 7v1v2 + (7/2)v2^2$ on $\alpha + Z^2$ (rev8 l.134), i.e. the full theta-lift exponent per label. A binary indefinite form can hit more than 3 classes mod 8. A next package would preregister "the uniform 140-label pool of the scaled Q(v) mod 8 (scale = least integer making Q integral on all $\alpha + Z^2$)", with natural density on a box and the cone weight w(v) as written, vs ρ_{res}. Its density convention must be fixed in PREREG. Not done here (stop rule).